

Revised Long-Memory Limit Theorems for the Hurst Function Estimator

Machine-checked against the Lean 4 formalization (repo `hurst_function_estimation`, closure commit `16715db`, acceptance `e76475a`)

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Scope. This note states the *revised* form of the long-memory branch of Theorem 3.3(iii) and Theorem 3.4 of the source manuscript (`19-A0S1825.pdf`), as it is now fully formalized in the accompanying Lean 4 project. Every condition and every clause of Theorem 2 below is machine-checked; §4 maps each element to its Lean theorem, and §5 lists, without softening, every deviation from the manuscript’s original assumptions and from the intermediate repair plan (`direct_proofs` files 22/24).

1 Setup

Let $f \in \Sigma(p, M)$ be a Hölder-regular Hurst profile on $(0, 1)$ in the sense of the source paper, let $t \in (0, 1)$ be a fixed interior point, and write

$$h = f(t), \quad \psi = 2 - 2h \in (0, \tfrac{1}{2}), \quad c = h(2h - 1) > 0.$$

The data are the centered Gaussian grid features of the harmonizable-type model with profile f ; the estimator is the local-polynomial log-square statistic with weights $w_{n,i}$ of order r (degree $r - 1$ design, $\sum_i w_{n,i} = 1$), bandwidth $\delta_n = n^{-\gamma}$, active scale $S_n = n\delta_n$, and the clipped calibration

$$\hat{G}_n = \sum_i w_{n,i} \log(Y_{n,i}^2), \quad \hat{H}_n = T_n \left(\frac{c_\sigma - \hat{G}_n}{2 \log n} \right),$$

where $Y_{n,i}$ are the standardized first-stride increment features (unit known scale $\sigma = 1$) and T_n clips its argument to $[0, 1]$; where T_n clips to $[0, 1]$ and $c_\sigma = \mathbb{E} \log Z^2$ with $Z \sim N(0, 1)$ (the pinned center; in the formalization this constant is `gaussianLogSquareMean`).

Let ω_r be the equivalent kernel of the local polynomial fit and let K be the Hilbert–Schmidt operator of the (unweighted, exponent- ψ , coefficient- c) Riesz kernel on L^2 . Define the compact self-adjoint operator

$$B = K^{1/2} M_{\omega_r} K^{1/2},$$

and let $\lambda = (\lambda_j)_{j \geq 0}$ be its eigenvalue sequence retaining every nonzero eigenvalue with multiplicity, ordered with positive and negative parts separately non-increasing and interleaved.

Condition 1 (Revised assumptions).

(R1) $p \geq 1$; $f \in \Sigma(p, M)$ with constant M .

(R2) $\frac{1}{2} < a \leq f(x) \leq b < 1$ for all $x \in (0, 1)$.

(R3) Long memory at the estimation point: $h = f(t) > \frac{3}{4}$.

(R4) Bandwidth compatibility (the b -band): $(1 - h)^2 < 1 - b$; equivalently, at $\gamma = h$, $(1 - \gamma)\psi < 2 - 2b$.

(R5) Bandwidth pinned at the feasible point $\gamma = h$, i.e. $\delta_n = n^{-h}$ (so $S_n = n^{1-h}$).

(R6) Fixed local-polynomial order $r \geq 0$.

The band (R4) is nonempty jointly with (R2)–(R3): one needs $h \leq b < 1 - (1 - h)^2$, and $h < 1 - (1 - h)^2$ holds for every $h \in (0, 1)$.

2 Main theorem

Theorem 2 (Revised Theorem 3.3(iii) + 3.4, long-memory branch). *Under (R1)–(R6), with Q the centered second Gaussian chaos*

$$Q = \sum_{j \geq 0} \lambda_j (Z_j^2 - 1), \quad (Z_j)_{j \geq 0} \stackrel{iid}{\sim} N(0, 1),$$

(the series converging in L^2 ; $\text{Var } Q = 2 \sum_j \lambda_j^2$), the truth-centered estimator satisfies

$$2 S_n^\psi \log n (\hat{H}_n - h) \implies -Q.$$

The sign is $-Q$: the reflection relative to the naive inversion is a substantive part of the repair (it changes the law whenever Q is asymmetric), and it is retained here exactly as in the corrected mathematics (*direct_proofs/14*).

Corollary 3 (No tuning constants). *Under (R1)–(R4), (R6) alone — with the near/far cut radius and every scalar rate parameter instantiated internally at the midpoint of the admissible window — the conclusion of Theorem 2 holds. The center constant c_σ and the projection band $d = (1 - h)/2$ are likewise internal.*

3 The honest rates behind the theorem

Proposition 4 (Drift decomposition; revised E5). *Under (R1)–(R6) the deterministic drift splits as*

$$2 S_n^\psi \log n |\mathbb{E} \hat{H}_n - h| \leq \underbrace{C_1 \frac{\log n}{n^{h-2(1-h)^2}}}_{\text{local-smoothing term}} + \underbrace{C_2 n^{-(2-2b)-2(1-h)^2}}_{\text{increment-mismatch term}} + \underbrace{C_3 \frac{1}{n^{2(1-h)^2} \log n}}_{\text{clipping term}},$$

where the first exponent is positive since $h > 3/4 > 2(1 - h)^2$, the second is positive exactly under the b -band (R4), and the third vanishes unconditionally. Moreover the increment-mismatch rate $n^{-(2-2b)}$ is attained exactly: no per-row expansion order reaches the rate n^{-h} , and consequently the manuscript's original bias envelope $|\mathbb{E} \hat{H}_n - h| \leq C n^{-h}$ fails for every non-constant profile with $f'(t) \neq 0$ (the window refutations are machine-checked; the lower-bound statement is derived by hand and registered).

Remark 5. Theorem 2 is therefore proved *through* the honest rates: the truth-centered limit needs only the drift to vanish, not an n^{-h} -rate bias envelope, and the drift condition reduces exactly to the b -band (R4).

4 Formal provenance

Every element of Theorem 2 is machine-checked; the chain, from bottom to top:

Element	Lean theorem	File
Internal Hilbert basis, closed spectrum	<code>exists_weightedRieszSpectrum_equiv_closed</code>	<code>WeightedRieszSpectrumClosed</code>
Cycle-integral identification ($k \geq 2$)	<code>diagSum_Bop_eq_weightedCycle</code>	<code>CompressionGeneralK</code>
Signed sorted spectrum keeps moments	<code>signedSortedSpectrum_hasSum_pow</code>	<code>GeneralSignedPowerMatching</code>
Row nondegeneracy (all n ; spectral)	<code>normalizedVaryingIncrement_ne_zero</code>	<code>FeatureRowNondegenerate</code>
Normalized energy $S^{2\psi-2} \sum \rho^2 = O(1)$	<code>hurstHolder_q1_actual_normalized_energy_*</code>	<code>NormalizedActualEnergy</code>
Fourth-order remainder $\rightarrow 0$	<code>hurstHolder_q1_actual_weighted_fourth_tendsto_zero</code>	<code>NormalizedActualRemainder</code>
Quadratic endpoint (constructed law)	<code>actualQ1LongStatistic_tendsto_secondChaos_generalSigned</code>	<code>ActualQ1SignedClosed</code>
Log layer (double-sum remainder)	<code>actualQ1_logStatistic_tendsto_secondChaos_doubleSum</code>	<code>P5LogHLayers</code>
Seam (Layer 1 $\rightarrow$ Layer 2)	<code>logStatistic_knownScaleH_seam_of_logLimit</code>	<code>P5SeamClosed</code>
Center band discharged ($d = (1 - h)/2$)	<code>centerBand_hband_discharged</code>	<code>CenterBandDischarge</code>
Scalar rates ($R = \lfloor n^\beta \rfloor$)	<code>feasibleR_*</code>	<code>FeasibleRates</code>
Second-order increment rate $n^{-(2-2b)}$	<code>varyingIncrement_norm_sq_secondOrder</code>	<code>IncrementLogSecondOrder</code>
Tail-based clipping transfer	<code>truncationCorrection_le</code>	<code>TruncationTransfer</code>
Unit variance rate $S^{-2\psi}$	<code>hurstHolder_q1_unitVariance_eventually_bounded</code>	<code>UnitVarianceRate</code>
Honest-rate lemma	<code>honestRate_drift_tendsto_zero</code>	<code>HonestRateDrift</code>
Main theorem	<code>actualQ1_knownScaleH_fullChain_honestRate</code>	<code>HonestRateClosure</code>
Refuted historical premises (audit)	<code>session3_fullChain_all_rows_impossible, session4_unnormalized_hE2_impossible, bias_window_*_infeasible</code>	<code>verification/</code>

Machine status. Lean 4.31.0 / mathlib v4.31.0; aggregate build 9223 jobs, exit 0; axiom audit of 2603 theorems, all within `{propext, Classical.choice, Quot.sound}`; zero sorry; automated re-verification via `verification/final_acceptance.py` (seven checks, all passed).

5 Deviations from the manuscript and the repair plan

1. **Sign.** The limit is $-Q$, not Q (substantive; matches the corrected mathematics of file 14, confirmed in Lean).
2. **Bandwidth system.** The manuscript’s original bandwidth/all-rows combination is unsatisfiable; both refutations are machine-checked. The revised window is (R4)–(R5), exactly the file-22 grid condition $(1 - \gamma)\psi < 2 - 2b$ instantiated at $\gamma = h$.
3. **Lower bound on a .** (R2) requires $a > \frac{1}{2}$, absent from the manuscript; it is needed by the second-order mixed-power machinery. Mild strengthening, satisfiable.

4. **γ pinned.** The honest-rate closure is assembled at the single feasible point $\gamma = h$; the manuscript's window allows other γ (the general- γ chain exists in the repo with a legacy bias premise). Coverage regions overlap but neither contains the other.
5. **Scale.** Known scale pinned at $\sigma = 1$ ($c_\sigma = \mathbb{E} \log Z^2$); the σ -rescaled known-scale variant and the unknown-scale extension (file 24, E6–E10) are not formalized.
6. **Honest bias rates.** The drift uses the attained rate $n^{-(2-2b)}$ (Proposition 4); the plan's presumed faster decay ($n^{-(2-b)}$, then $n^{-\gamma}$ envelopes) was refuted during formalization and is registered, not papered over.
7. **Short-memory and critical branches** of Theorem 3.3, the minimax program, and all two-dimensional extensions are outside this closure and remain as separately registered work.